Experiment No. 2

Exploratory Data Analysis (EDA) on a Real-World Business Dataset using Python

Dataset used: Titanic Survival Dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

try:
    df = pd.read_csv('titanic_cleaned.csv')
except FileNotFoundError:
    df = sns.load_dataset('titanic').dropna(subset=['age', 'fare']).drop_dup

print("Dataset loaded. Shape:", df.shape)
df.head()
```

```
Dataset loaded. Shape: (775, 22)
```

	survived	pclass	sex	age	sibsp	parch	fare	class	who	adult_
0	0	3	male	22.0	1	0	7.2500	Third	man	
1	1	1	female	38.0	1	0	71.2833	First	woman	F
2	1	3	female	26.0	0	0	7.9250	Third	woman	F
3	1	1	female	35.0	1	0	53.1000	First	woman	F
4	0	3	male	35.0	0	0	8.0500	Third	man	

5 rows × 22 columns

```
print("Shape (rows, columns):", df.shape)
print("\nColumn names:")
print(df.columns.tolist())
print("\nData types:")
print(df.dtypes)
```

```
Shape (rows, columns): (775, 22)

Column names:
['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare', 'class', 'who'
```

```
Data types:
survived           int64
pclass             int64
sex               object
age              float64
sibsp              int64
parch              int64
fare             float64
class             object
who               object
adult_male          bool
alive             object
alone               bool
sex_encoded        int64
alive_encoded      int64
embarked_C          bool
embarked_Q          bool
embarked_S          bool
age_scaled       float64
fare_scaled      float64
family_size        int64
age_group         object
fare_category     object
dtype: object
```

```
print("Summary statistics for numerical columns:")
df.describe()
```

Summary statistics for numerical columns:

	survived	pclass	age	sibsp	parch	fare	s
count	775.000000	775.000000	775.000000	775.000000	775.000000	775.000000	
mean	0.412903	2.246452	29.363123	0.529032	0.420645	26.593137	
std	0.492674	0.853574	13.211598	0.990326	0.840565	22.873099	
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000	
25%	0.000000	1.000000	21.000000	0.000000	0.000000	8.050000	
50%	0.000000	3.000000	28.000000	0.000000	0.000000	15.900000	
75%	1.000000	3.000000	36.000000	1.000000	1.000000	34.197900	
max	1.000000	3.000000	58.500000	8.000000	6.000000	73.419750	

```
print("Value counts for 'sex':")
print(df['sex'].value_counts())

print("\nValue counts for 'class':")
print(df['class'].value_counts())

print("\nValue counts for 'survived' (0 = No, 1 = Yes):")
print(df['survived'].value_counts())
```

```
Value counts for 'sex':
sex
male      483
female    292
Name: count, dtype: int64

Value counts for 'class':
class
Third     401
First     210
Second    164
Name: count, dtype: int64

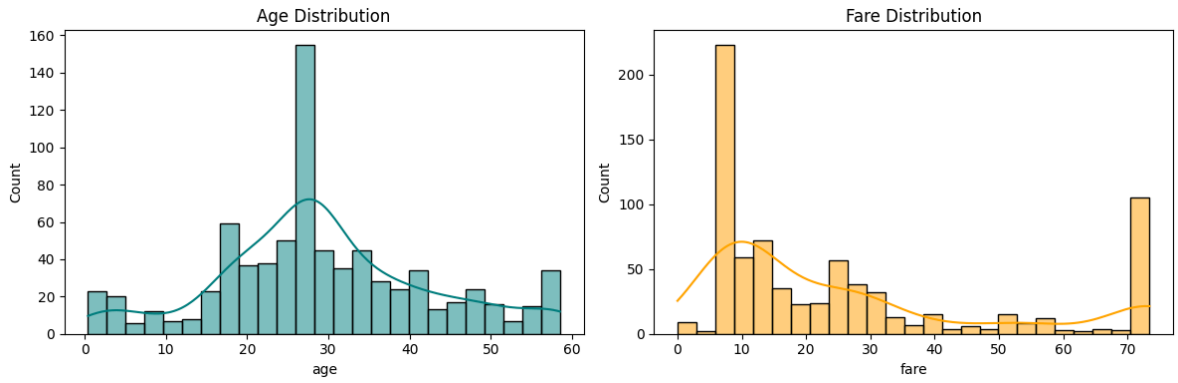
Value counts for 'survived' (0 = No, 1 = Yes):
survived
0    455
1    320
Name: count, dtype: int64
```

```
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

sns.histplot(df['age'], kde=True, bins=25, ax=axes[0], color='teal')
axes[0].set_title('Age Distribution')

sns.histplot(df['fare'], kde=True, bins=25, ax=axes[1], color='orange')
axes[1].set_title('Fare Distribution')

plt.tight_layout()
plt.savefig('eda_histograms.png')
plt.show()
```




```
fig, axes = plt.subplots(1, 3, figsize=(15, 4))

sns.countplot(x='sex', data=df, ax=axes[0], palette='Set2')
axes[0].set_title('Passenger Count by Sex')

sns.countplot(x='class', data=df, ax=axes[1], palette='Set2')
axes[1].set_title('Passenger Count by Class')

sns.countplot(x='survived', data=df, ax=axes[2], palette='Set2')
axes[2].set_title('Survived (0=No, 1=Yes)')

plt.tight_layout()
```

```
plt.savefig('eda_categorical_counts.png')
plt.show()
```

```
/tmp/ipykernel_2605/695283444.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed i

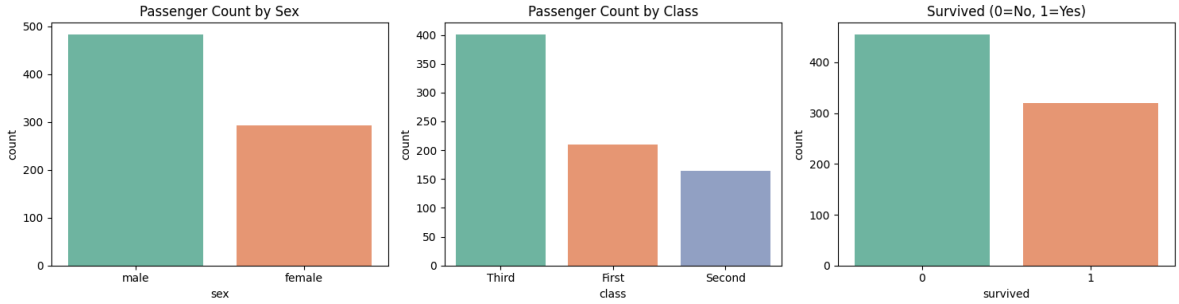
  sns.countplot(x='sex', data=df, ax=axes[0], palette='Set2')
/tmp/ipykernel_2605/695283444.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed i

  sns.countplot(x='class', data=df, ax=axes[1], palette='Set2')
/tmp/ipykernel_2605/695283444.py:9: FutureWarning:

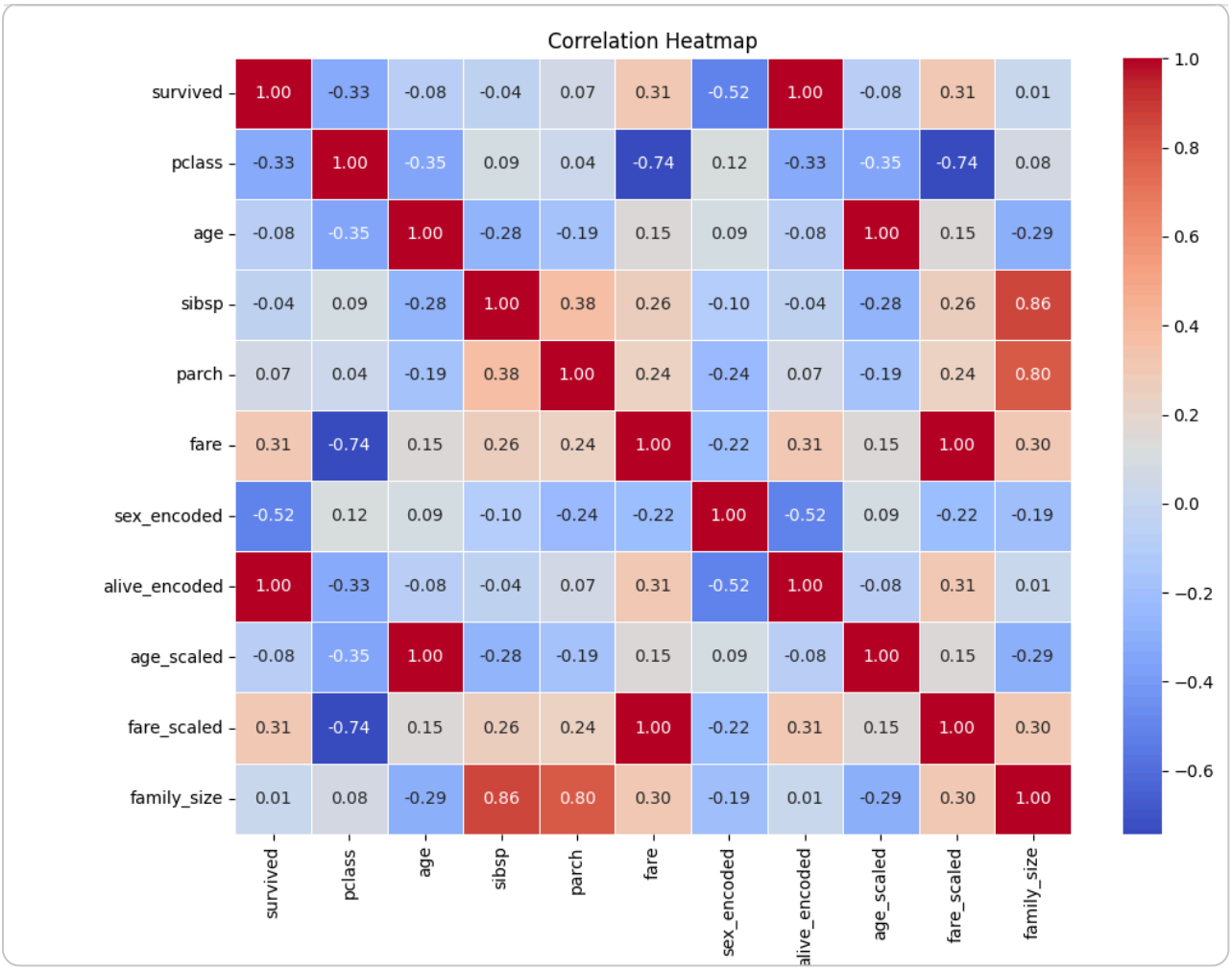
Passing `palette` without assigning `hue` is deprecated and will be removed i

  sns.countplot(x='survived', data=df, ax=axes[2], palette='Set2')
```




```
numeric_df = df.select_dtypes(include=[np.number])
corr_matrix = numeric_df.corr()

plt.figure(figsize=(10, 8))
sns.heatmap(corr_matrix, annot=True, fmt='.2f', cmap='coolwarm', linewidths=
plt.title('Correlation Heatmap')
plt.tight_layout()
plt.savefig('eda_correlation_heatmap.png')
plt.show()
```




```
fig, axes = plt.subplots(1, 2, figsize=(12, 4))

sns.boxplot(x='class', y='fare', data=df, ax=axes[0], palette='Set3')
axes[0].set_title('Fare by Class (Box Plot)')

sns.boxplot(x='survived', y='age', data=df, ax=axes[1], palette='Set3')
axes[1].set_title('Age by Survival (Box Plot)')

plt.tight_layout()
plt.savefig('eda_boxplots.png')
plt.show()
```

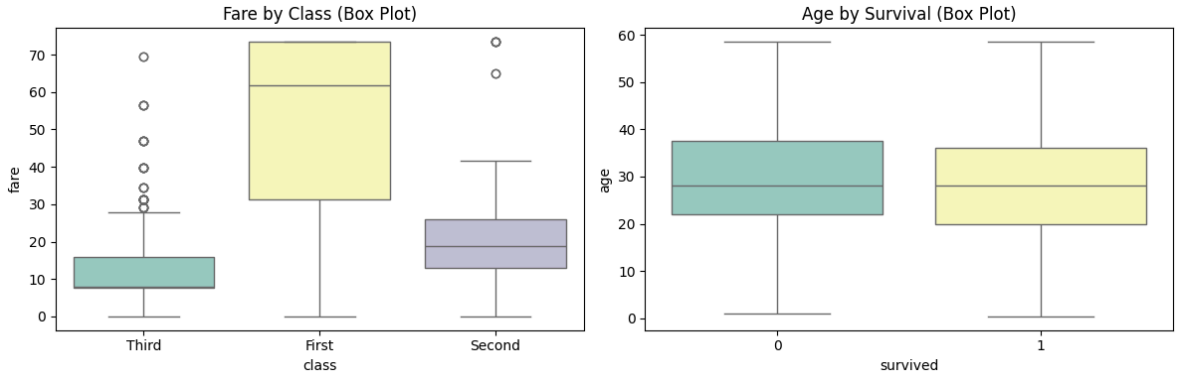
```
/tmp/ipykernel_2605/1847731207.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed i

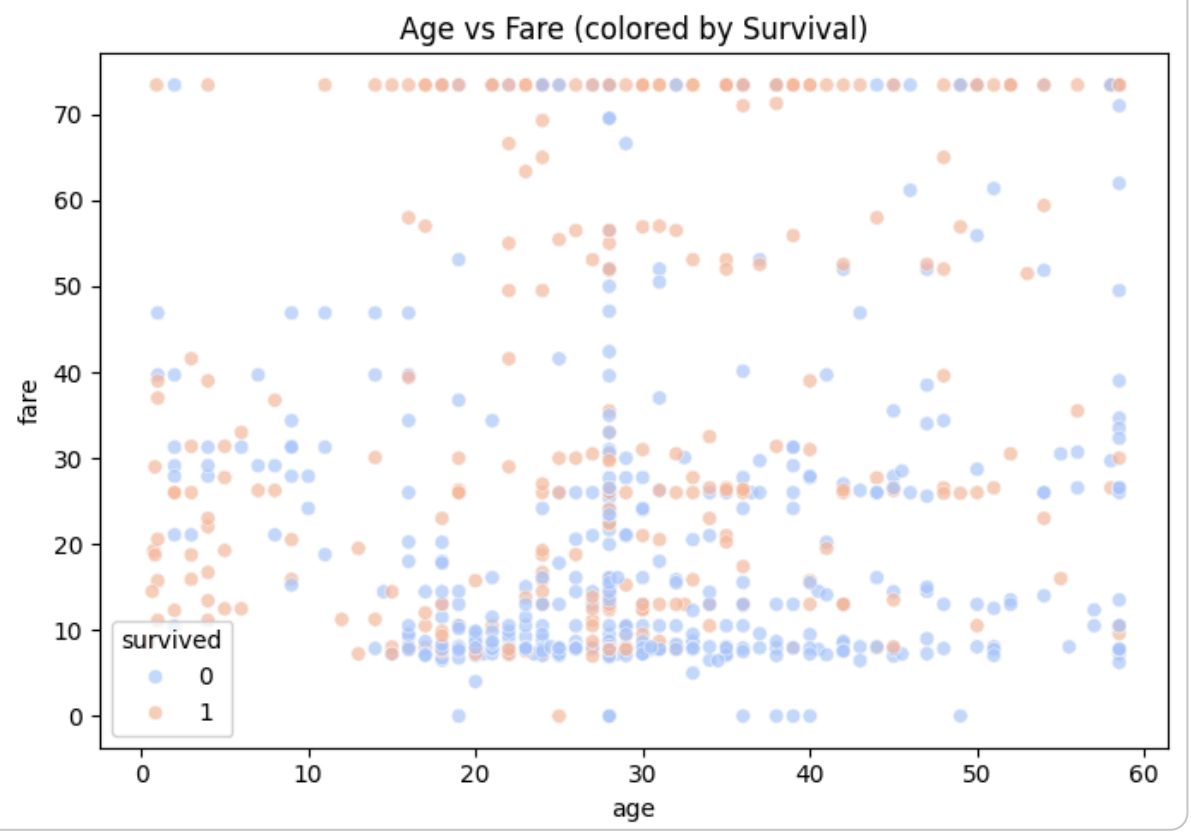
  sns.boxplot(x='class', y='fare', data=df, ax=axes[0], palette='Set3')
/tmp/ipykernel_2605/1847731207.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed i

  sns.boxplot(x='survived', y='age', data=df, ax=axes[1], palette='Set3')
```




```
plt.figure(figsize=(7, 5))
sns.scatterplot(x='age', y='fare', hue='survived', data=df, palette='coolwar
plt.title('Age vs Fare (colored by Survival)')
plt.tight_layout()
plt.savefig('eda_scatterplot.png')
plt.show()
```




```
print("Univariate - Age summary:")
print(df['age'].describe())
```

```
Univariate - Age summary:
count    775.000000
mean      29.363123
std       13.211598
min        0.420000
25%       21.000000
50%       28.000000
75%       36.000000
max       58.500000
Name: age, dtype: float64
```

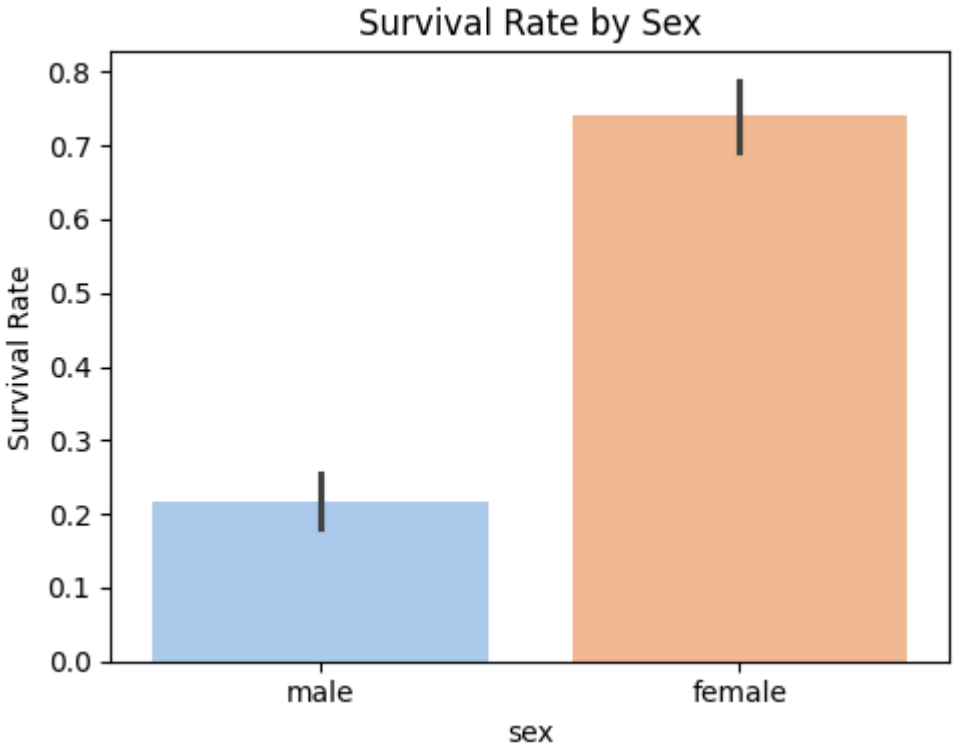
```
print("Bivariate - Survival rate by Sex:")
print(df.groupby('sex')['survived'].mean())

plt.figure(figsize=(5, 4))
sns.barplot(x='sex', y='survived', data=df, palette='pastel')
plt.title('Survival Rate by Sex')
plt.ylabel('Survival Rate')
plt.tight_layout()
plt.savefig('eda_bivariate_sex_survival.png')
plt.show()
```

```
Bivariate - Survival rate by Sex:
sex
female    0.739726
male      0.215321
Name: survived, dtype: float64
/tmp/ipykernel_2605/4193866006.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed i

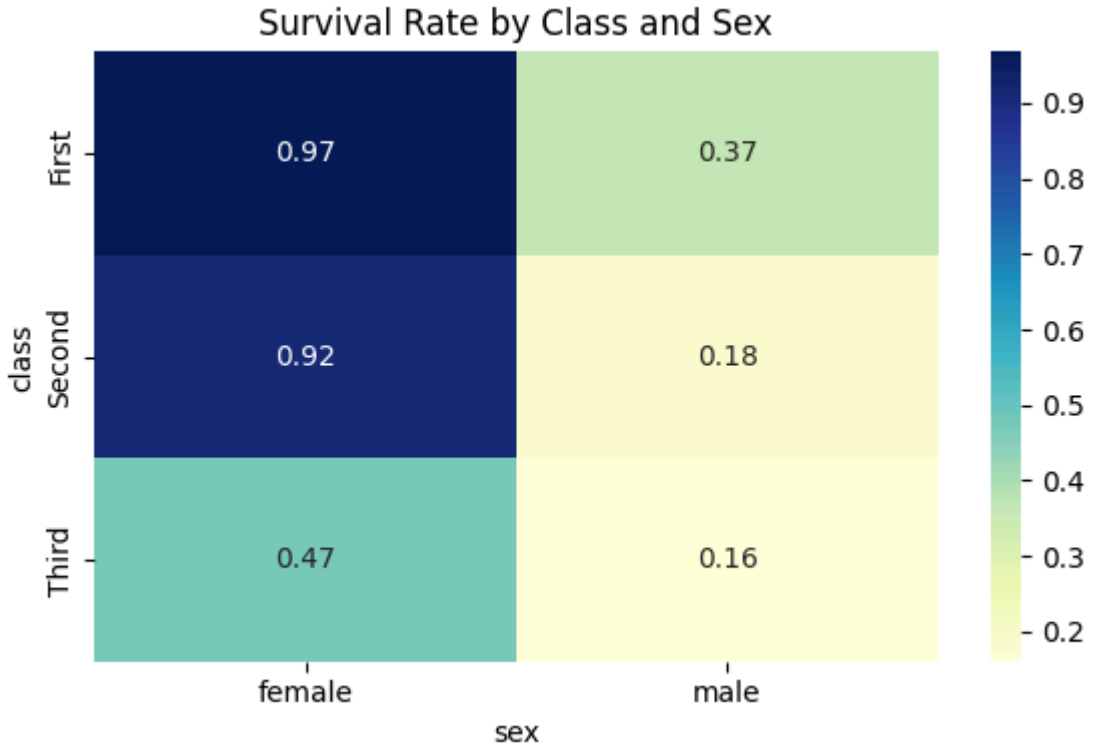
  sns.barplot(x='sex', y='survived', data=df, palette='pastel')
```



```
print("Multivariate - Survival rate by Class and Sex:")
pivot = df.pivot_table(index='class', columns='sex', values='survived',
print(pivot)

plt.figure(figsize=(6, 4))
sns.heatmap(pivot, annot=True, cmap='YlGnBu', fmt='.2f')
plt.title('Survival Rate by Class and Sex')
plt.tight_layout()
plt.savefig('eda_multivariate_heatmap.png')
plt.show()
```

```
Multivariate - Survival rate by Class and Sex:
sex       female      male
class
First   0.967742  0.367521
Second  0.916667  0.184783
Third   0.472441  0.160584
```




```
female_survival = df[df['sex'] == 'female']['survived'].mean()
male_survival = df[df['sex'] == 'male']['survived'].mean()
first_class_survival = df[df['class'] == 'First']['survived'].mean()
third_class_survival = df[df['class'] == 'Third']['survived'].mean()
age_fare_corr = df['age'].corr(df['fare'])

print("KEY OBSERVATIONS")
print("-" * 40)
print(f"1. Female survival rate: {female_survival:.2%}")
print(f"2. Male survival rate: {male_survival:.2%}")
print(f"3. First class survival rate: {first_class_survival:.2%}")
```